Draft version March 14, 2026
Typeset using LaTeX **twocolumn** style in AASTeX631

Measuring Planck's Constant using Photoelectric Effect

Songhang Li

(Dated: October 2022)

1. RESULTS AND DISCUSSION

The experiment result shows that the stop voltage is depend on the wavelength(frequency) of the light but is independent on the intensity of the light, which is matches the xxx theory. The data consists of different light colors and different orders of the spectrum to vary the wavelength and the intensity of the light respectively. The slope of the linear equation is xxx which is h/e. The Planck's constant measured in this experiment is h=slope × e =xxx.

1.1. *Measurement and Data*

–

Light	Wavelength	Frequency	V_s
1^{st} order	nm	Hz	V
UV	365.5	8.208×10^{14}	1.837
Vio	404.7	7.413×10^{14}	1.556
Blue	435.8	6.884×10^{14}	1.422
Green	546.1	5.493×10^{14}	0.760
Yellow	578.0	5.190×10^{14}	0.675
2^{nd} order	Wavelength	Frequency	V_s
UV	365.5	8.208×10^{14}	1.802
Vio	404.7	7.413×10^{14}	1.566
Blue	435.8	6.884×10^{14}	1.403
Green	546.1	5.493×10^{14}	0.746
Yellow	578.0	5.190×10^{14}	0.506

There are 11 different measurements in the different light-travel distances that were eventually measured and presented in Table 1. In Table 1, the number of experiment trials, the light-travel distance (in the unit of cm), and the time delay of the signal (in the unit of ns) are listed in the first, second, and the third column, respectively. During the experiment, the setup and the results accordingly were measured twice so that the value was given by the average of two measurements and the uncertainty was given by the standard deviation of both measurements. The measurement of the light-travel distance is fairly more accurate than the one of the time-delay of the signal – most of the uncertainties of those measurements are within about 1 centimeter except for the first and the seventh. Most of the uncertainty of the light-travel distance was contributed by the hardness in aligning the measuring tape with the laser and mirror; in addition, there are a few uncertainties (±0.1 cm) caused by measuring the distance from the laser source and the receiver. There are great uncertainties in the signal delay measurements. The uncertainties of the time delay of the signal resulted from the difficulty of identifying the fuzzy contour of the sine waveform.

Table 1. Measurements obtained from P216 Lab

#	$d_{\rm light}$	Δt
–	cm	ns
1	402.9 ± 5.9	11.0 ± 7.1
2	585.4 ± 0.5	20.0 ± 5.7
3	578.1 ± 0.4	23.0 ± 9.9
4	1047.1 ± 0.4	30.0 ± 8.5
5	1426.9 ± 0.4	41.0 ± 12.7
6	1536.1 ± 1.4	48.0 ± 5.7
7	2289.6 ± 3.0	68.0 ± 5.7
8	1778.4 ± 1.0	53.0 ± 9.9
9	2074.7 ± 0.9	66.0 ± 0
10	2211.5 ± 1.1	63.0 ± 12.7
11	1547.4 ± 0.5	54.0 ± 5.7

Note—There is no uncertainty for Δt given in the ninth trial of this experiment, since two measurements were giving the same value.

Those measurements are showing a nice linear relationship, which can be indicated by the scatter plot in Figure 1. In the scatter plot, the light-travel distance and the time delay of the signal are shown on the x- and y-axis, respectively. In addition to the data points, a best-fit line was also shown in the figure in red. Despite the uncertainties, those data points and the best-fit line suggest that $D_{\rm light}$ and Δt are strongly correlated. Those data points appear in such tread since as the light-

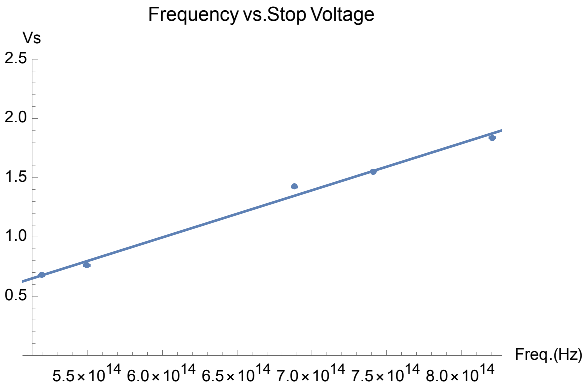

Figure 1. The scatter plot of the time-delay of the signal vs. light-travel distance. In this figure, lab measurements and the curve-fitted result are indicated in blue points and a red line, respectively. The uncertainty in each data point is represented by error bars in both the x and y directions, although the distance uncertainty in the y direction is too small to be noticed.

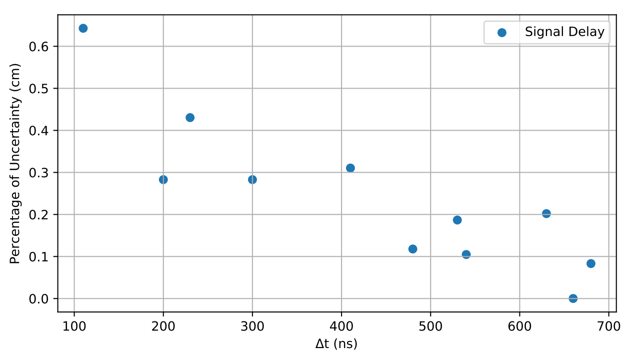

Figure 2. Percentage of the uncertainty in signal delay measurements, indicating that uncertainties get lower as the signal delay gets longer.

travel distance gets longer, the time that the light spends passing through is getting longer. The fitted equation of those data points is given as the following:

$$\Delta\mathrm{t} = [3.13_{\pm 0.23} \times 10^{-2}] \times \mathrm{D_{light}} + 1.13_{\pm 4.82}$$

Even though the constant in the linear equation has expected to be zero, it has a value of about 1 ns. This unexpected phenomenon is due to the systematic error caused by the measurement devices and signal cables. The speed of light has expected to be $\mathrm{c} = \frac{\mathrm{D_{light}}}{\Delta\mathrm{t}}$, which is the inverse of the slope. Eventually, the speed of light is given as $(3.20 \pm 0.24) \times 10^8\ \mathrm{m \cdot s^{-1}}$.

2. DISCUSSION

2.1. *Great Uncertainties in Signal Delay Measurements*

There are great uncertainties in the signal delay measurements in this work – the uncertainties of measurements are up to 9.9 and 12.7 in the third and fifth trials, which is 43% and 30% of the value itself. As those uncertainties are further analyzed, it has been found that as the signal delay gets larger, we are getting more confident in those measurements. This can be inferred from Fig. 2, in which the delay of the signal and the percentage of its uncertainty (given as $\frac{\delta(\Delta\mathrm{t})}{\Delta\mathrm{t}}$) has shown on the x- and y-axis. As the Δt gets longer, the percentage of uncertainty has getting lower.

This trend can be explained by the mechanism of how the uncertainty arises. The uncertainty in the signal delay was caused by the hardness of identifying the fuzzy contour of the sine waveform, while the fuzzy contour was mostly due to the noise both in the environment and caused by the instruments. The uncertainty caused by those factors would remain constant as the time of the signal delay gets longer. Therefore, as the signal delay gets larger and the uncertainty remains constant, the percentage of the uncertainty gets lower.

Inspired by this phenomenon, the time delay measurements can be measured more accurately by reproducing the experiment in a longer light-travel distance, which will result in a longer delay of signal and a lower percentage of uncertainty. (Should be noticed that the experiment should be conducted more carefully when $\Delta\mathrm{t} > \mathrm{T}$, where T is the period of the sine wave.)